

DATA STRUCTURES

Data Structures & Algorithms: Design and Complexity Analysis

Arrays, Linked Lists, Stacks, Queues, Binary Trees, Heaps, Graphs, Hashing & Asymptotic Complexity

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CHAPTER 1

Asymptotic Analysis, Arrays & Linked Data Structures

1. Theoretical Foundations & Scope

Choosing the right data structure directly impacts software runtime and memory footprint. This chapter covers Big-O, Big-Omega, Big-Theta notations, contiguous arrays, singly/doubly linked lists, and circular lists.

2. Core Conceptual Framework

- **Asymptotic Complexity:** Characterizing worst-case, best-case, and average-case runtimes as input size approaches infinity.
- **Arrays vs Linked Lists:** $O(1)$ random access vs $O(n)$ access with $O(1)$ pointer-based insertions without resizing.
- **Doubly & Circular Lists:** Bi-directional pointer traversal and continuous round-robin buffer representations.

Key Practical Insights & Examination Takeaways:

- ' Prove algorithmic complexity using formal mathematical limits.
- ' Implement pointer manipulation algorithms without memory leaks.
- ' Choose between contiguous memory cache locality and dynamic link node allocation.

CHAPTER 2

Stacks, Queues, Recursion & Hash Tables

1. Theoretical Foundations & Scope

Stacks and queues serve as fundamental abstract data structures in operating systems and parsers. We examine LIFO stack mechanics, FIFO circular queues, priority queues, and hash table collision resolution techniques.

2. Core Conceptual Framework

- **Stack Applications:** Function call call-stacks, balanced parenthesis checking, and infix to postfix expression conversion.
- **Circular Queues:** Ring buffer implementations using modulo arithmetic preventing false queue-full conditions.
- **Hash Collision Resolution:** Separate chaining with linked lists versus open addressing (linear probing, quadratic probing, double hashing).

Key Practical Insights & Examination Takeaways:

- ' Evaluate postfix expressions using an operand stack in $O(n)$ time.
- ' Analyze load factor effects on hash table search complexity.
- ' Differentiate priority queues implemented via arrays versus binary heaps.

CHAPTER 3

Hierarchical Structures: Binary Trees, Heaps & Graph Traversal

1. Theoretical Foundations & Scope

Non-linear structures represent complex real-world relationships. This chapter covers Binary Search Trees (BST), AVL self-balancing trees, binary max/min heaps, graph representations (adjacency matrix vs list), and BFS/DFS traversal.

2. Core Conceptual Framework

- **Binary Search Trees & AVL:** In-order traversal yielding sorted keys, and tree rotations maintaining $O(\log n)$ height balance.
- **Binary Heaps:** Complete binary tree representations stored in flat arrays satisfying heap-order invariants.
- **Graph Traversals (BFS & DFS):** Queue-based Breadth-First Search for shortest paths versus stack/recursion-based Depth-First Search.

Key Practical Insights & Examination Takeaways:

- ' Execute single and double AVL tree rotations following node insertions.
- ' Perform heapify operations to construct heaps in linear $O(n)$ time.
- ' Implement Dijkstra and Prim algorithms using adjacency lists and priority queues.

Academic References & Further Study

[1] Quality Education Academic Standards & Pedagogical Guidelines, 2026 Edition.

[2] International Computer Science & Engineering Curriculum Framework.

[3] Educational Technology & Assessment Methodology Research Series.

[4] Open Access Educational Resource Repository - Quality Education Platform.

